

BIOST 2155: Assignment 2

YOUR NAME HERE (YOUR PITT EMAIL HERE)

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Instructions

- This file serves as a template for your assignment. **Remember to set `eval = TRUE` below when compiling the report.**

```
knitr::opts_chunk$set(echo = TRUE, eval = FALSE,
  fig.height = 4, fig.width = 7, fig.align = "center",
  message=FALSE)
```

- Use comments to annotate and explain your code whenever possible.
- This assignment covers **Regression II** (non-linear regression, splines, kernel trick) and **Classification I** (logistic regression, generative models). It uses two data sets: **Boston Housing** (`assign1_regData.csv`, continued from Assignment 1) for Part A, and **Pima Diabetes** (`assign2_diabetes.csv`) for Part B.

Theory (12 points)

T1. Continuity and smoothness of cubic splines (12 points)

ISLR Problem 7.9.1, page 321. A cubic regression spline with one knot at ξ can be obtained using a basis of the form $x, x^2, x^3, (x - \xi)_+^3$, where $(x - \xi)_+^3 = (x - \xi)^3$ if $x > \xi$ and equals 0 otherwise. We will now show that a function of the form

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x - \xi)_+^3$$

is indeed a cubic regression spline, regardless of the values of $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$.

- Find a cubic polynomial $f_1(x) = a_1 + b_1 x + c_1 x^2 + d_1 x^3$ such that $f(x) = f_1(x)$ for all $x \leq \xi$. Express a_1, b_1, c_1, d_1 in terms of $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$.
- Find a cubic polynomial $f_2(x) = a_2 + b_2 x + c_2 x^2 + d_2 x^3$ such that $f(x) = f_2(x)$ for all $x > \xi$. Express a_2, b_2, c_2, d_2 in terms of $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$. We have now established that $f(x)$ is a piecewise polynomial.
- Show that $f_1(\xi) = f_2(\xi)$, i.e. that $f(x)$ is continuous at ξ .
- Show that $f'_1(\xi) = f'_2(\xi)$, i.e. that $f'(x)$ is continuous at ξ .
- Show that $f''_1(\xi) = f''_2(\xi)$, i.e. that $f''(x)$ is continuous at ξ .

Together, (c), (d), and (e) establish that $f(x)$ is indeed a cubic spline.

Application, Part A: Regression II — Non-linear modeling (38 points)

Dataset: Boston Housing, saved as `assign1_regData.csv` (same data as Assignment 1).

```
# Load required libraries and dataset
pacman::p_load(tidyverse, caret, splines, gam)
set.seed(42)
data <- read_csv("assign1_regData.csv") |> drop_na()
data$CHAS <- as.factor(data$CHAS)

train_index <- createDataPartition(data$MEDV, p = 0.8, list = FALSE)
train <- data[train_index, ]
test <- data[-train_index, ]
```

B1. Polynomial Regression (5 + 5 + 5 = 15 points)

1. Consider a polynomial regression with `LSTAT` as the only predictor. Use 10-fold CV on the training data to select the optimal polynomial degree (1 to 10). Plot how CV error varies with polynomial degree.

```
# your code here
```

2. Report the optimal degree and the test set RMSE for that model.

```
# your code here
```

3. Plot the fitted polynomial curve over a scatterplot of `MEDV` vs `LSTAT`. Does the relationship look like it needed a non-linear fit, or would a straight line have done nearly as well?

```
# your code here
```

B2. Splines (5 + 5 = 10 points)

1. Fit a natural spline (natural spline = cubic spline + linearity constraint in the tails) with `RM` as the predictor, using 4 degrees of freedom. Report the test set RMSE.

```
# your code here
```

2. Compare this test RMSE to a simple linear fit of `MEDV` on `RM` alone. Did the spline improve prediction? Briefly explain, in terms of the shape of the `RM-MEDV` relationship, why or why not.

```
# your code here
```

B3. Generalized Additive Model (GAM) (8 + 5 = 13 points)

1. Fit a GAM for `MEDV` using natural splines for both `LSTAT` and `RM` (4 df each), plus `CHAS` as a linear term. Report the test set RMSE.

your code here

2. Compare the test RMSE of this GAM to the polynomial model from B1 and the spline model from B2. Which model would you recommend, and why?

your code here

Application, Part B: Classification I — Logistic regression and LDA (40 points)

Dataset: Pima Diabetes, saved as `assign2_diabetes.csv`.

This is a classic dataset of $n = 768$ women of Pima Indian heritage, used to predict the onset of diabetes within five years based on diagnostic measurements. Your goal is to build and compare two classifiers: logistic regression (a discriminative model) and linear discriminant analysis (a generative model). **Please set a random seed (`set.seed(42)`) before any step involving data splitting or randomization.**

- **Pregnancies:** Number of times pregnant
- **Glucose:** Plasma glucose concentration (2-hour oral glucose tolerance test)
- **BloodPressure:** Diastolic blood pressure (mm Hg)
- **SkinThickness:** Triceps skin fold thickness (mm)
- **Insulin:** 2-Hour serum insulin (μ U/mL)
- **BMI:** Body mass index
- **DiabetesPedigreeFunction:** A function that scores likelihood of diabetes based on family history
- **Age:** Age in years
- **Outcome:** 1 if the patient developed diabetes, 0 otherwise

```
pacman::p_load(tidyverse, caret, MASS, pROC)
set.seed(42)
diab <- read_csv("assign2_diabetes.csv") |> drop_na()
diab$Outcome <- factor(diab$Outcome, levels = c(0, 1), labels = c("No", "Yes"))
```

C1. Data Exploration & Splitting (5 + 5 = 10 points)

1. Use `summary()` to explore the dataset. Comment on any variables that look like they may have implausible zero values (e.g. `Glucose` or `BMI` of zero) and briefly explain why that matters for the analysis (you do not need to fix it — just note the issue).

your code here

2. Randomly split the data into a training set (75%) and a test set (25%), stratified by `Outcome` (use `caret::createDataPartition, seed = 42`).

your code here

C2. Logistic Regression (5 + 5 + 5 = 15 points)

1. Fit a logistic regression model predicting `Outcome` from all eight predictors on the training data. Which predictors are statistically significant at the 0.05 level?

your code here

- Using a 0.5 probability threshold, generate class predictions on the test set and report the confusion matrix (`caret::confusionMatrix`).

your code here

- Plot the ROC curve for this model on the test set and report the AUC.

your code here

C3. Linear Discriminant Analysis & Comparison (5 + 5 + 5 = 15 points)

- Fit an LDA model with the same eight predictors on the training data, and generate class predictions on the test set.

your code here

- Report the LDA confusion matrix and AUC (using the posterior probability of **Yes**) on the test set.

your code here

- Interpretation: *Compare logistic regression and LDA on accuracy, sensitivity, specificity, and AUC. LDA assumes predictors are approximately multivariate normal within each class and share a common covariance matrix — do you expect that assumption to hold reasonably well for this data? Which model would you recommend for this problem, and why?*